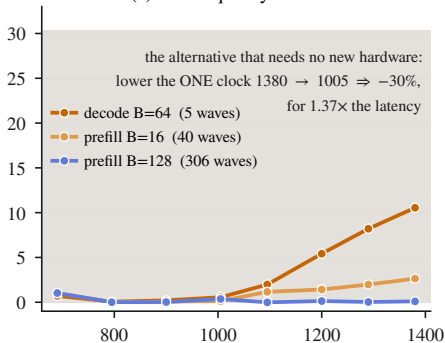


energy saved vs. grouped GEMM

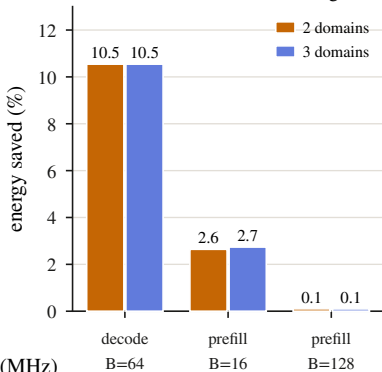
at the same latency (%)

(a) two frequency domains



SLO tightness: clock the baseline grouped GEMM must run at (MHz)

(b) at the tightest SLO (1380 MHz):  
a third domain adds nothing



PREDICTED, not measured. Tiles per expert =  $\lceil n_e/128 \rceil \times \lceil 1024/128 \rceil$  from the measured OLMoE routing; a domain of  $s_d$  SMs holding  $N_d$  tiles takes  $\lceil N_d/s_d \rceil$  waves, so its clock is forced to the lowest 15 MHz grid point meeting the deadline; energy =  $\sum_d [N_d \kappa(f_d) + s_d P_{\text{leak}}(f_d) T]$  with  $\kappa$  and  $P_{\text{leak}}$  measured.

Search covers every contiguous grouping of load-sorted experts and every SM allocation. The optimum always puts the LIGHT experts on the low-frequency domain.